

Testing of insulating oils

Detection of corrosive sulfur Silver strip test

DIN
51 353Prüfung von Isolierölen; Prüfung auf korrosiven Schwefel;
Silberstreifenprüfung

Supersedes September 1977 edition.

*In keeping with current practice in standards published by the International Organization for Standardization (ISO),
a comma has been used throughout as the decimal marker.*

1 Field of application

This standard applies to insulating oils as specified in DIN VDE 0370 Part 1 and DIN VDE 0370 Part 2.

2 Scope

This standard specifies a method of detecting such amounts of corrosive sulfur as would interfere in the operation of electrical apparatus.

3 Concept

For the purpose of this standard, sulfur and sulfur compounds are classified as corrosive if, under the conditions of this test method, they cause visible discolouration of a silver strip by forming silver sulfide as specified in sub-clause 11.2.

4 Principle

A silver strip is kept in the insulating oil for 18 hours at 100°C and is then inspected for discolouration of the surface.

5 Designation

Designation of the silver strip method of determining corrosive sulfur in insulating oils (S):

Test DIN 51 353 - S

6 Apparatus

The following apparatus shall be used:

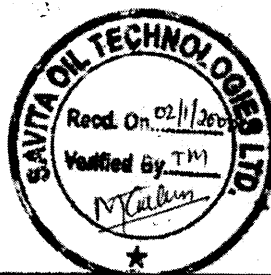
heating cabinet as specified in DIN 50011 Part 1, capable of being maintained at $(100 \pm 2)^\circ\text{C}$;
100 ml conical flasks fitted with DIN 12 242 - KNS 29/32 ground joint glass stoppers, e.g. DIN 12 387 - E 100 ANS 29 flasks, made of chemically resistant glass, e.g. DIN 12 111 glass, hydrolytic class 3.

7 Reagents and apparatus

The following reagents and apparatus shall be used:
chromosulfuric acid, prepared by dissolving 50 g of $\text{K}_2\text{Cr}_2\text{O}_7$ in 1 l of not less than 96% (m/m) sulfuric acid, H_2SO_4 , $\rho = 1,84 \text{ g/ml}$. The solution shall be kept stoppered in reagent bottles fitted with ground joint glass stoppers;
sulfur-free acetone (2-propanone);
sheet silver 1) with $w(\text{Ag})$ of 99,99% by mass, 0,5 mm thick, hard-rolled blank, free from blemishes.

1) Information on sources of supply can be obtained from: *DIN-Bezugsquellen für normgerechte Erzeugnisse im DIN*, Burggrafenstraße 6, D-1000 Berlin 30.

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Note. It is recommended that the sheet silver be kept in stock in convenient bands of 20 mm in width, from which the silver strips can be cut after polishing.

abrasive consisting of loose silicon carbide grains of grain size P 150 as specified in DIN 69 176 Part 2;

DIN 61 640 – A absorbent cotton wool for eye dressings or absorbent cotton wool for surgical dressings, e.g.

DIN 61 640 – V – BW absorbent cotton wool for surgical dressings;

filter paper for quantitative analysis, e.g. DIN 53 135 – 2B filter paper.

8 Sampling

Sampling shall be carried out in accordance with DIN 51 750 Part 1 and Part 2.

9 Preparation

9.1 Cleaning of flasks

Clean the flasks with fresh chromosulfuric acid and wash with tap water until all traces of the acid are removed. Then rinse the flasks three times with distilled water and dry them in a heating cabinet at 105 °C.

The stoppers shall be cleaned in a similar manner.

Note. Water purified by ion exchangers shall not be used, since it contains substances which may affect the appearance of the silver strip.

9.2 Preparation of silver strip

9.2.1 Prior to the test, polish the sheet silver by hand with light pressure being applied through the pad of absorbent cotton wool. For this purpose, pour the abrasive onto a clean glass plate and pick it up with the aid of a pad of absorbent cotton wool moistened with acetone. Carry out the polishing in the direction of the long axis of the silver strip to be cut from the sheet. Avoid circular motion. To avoid direct contact between the sheet and the hands or laboratory table, place a filter paper under the sheet.

9.2.2 When the surfaces on both sides of the sheet have been uniformly matted to a silk finish, terminate the polishing and without delay continue with the preparations and commence the test.

9.2.3 Using clean, degreased scissors, cut from the sheet a 20 mm × 40 mm strip and wipe it with fresh pads of

absorbent cotton wool dipped in acetone in order to free it from abrasive and metal dust.

9.2.4 Bend the silver strip to a V-shape, with its two limbs at an angle of approximately 60°. Take special care not to touch it by hand.

9.2.5 Wash the prepared silver strip once again with acetone. Then immediately transfer it to a heating cabinet maintained at $(100 \pm 2)^\circ\text{C}$, and place it edge-down on filter paper for 1 minute.

10 Procedure

Fill a flask cleaned as specified in subclause 9.1 with 100 ml of the insulating oil sample. Place the silver strip prepared as specified in subclause 9.2 on the bottom of the flask, resting on the edges of the two limbs, using cleaned tweezers. Lubricate the stopper with a small amount of the oil sample and place it loosely on the flask. Place the flask in the heating cabinet maintained at $(100 \pm 2)^\circ\text{C}$. Keep it in the cabinet for $18\text{ h} \pm 30\text{ min}$. Subsequently remove the silver strip from the oil sample using the tweezers and wash with acetone to remove the oil completely. Then inspect the surface of the silver strip in daylight at an angle of about 45°, avoiding bright sunlight, and examine it for discolouration by comparing it with a freshly treated silver strip.

11 Evaluation

11.1 An insulating oil may be regarded as being free from corrosive sulfur if it causes no recognizable discolouration of the silver strip under the conditions of this test method. Thermal stresses acting on the silver strip in the test can, however, give rise to a very weak golden yellow colouration. This should be disregarded.

11.2 Insulating oils containing corrosive sulfur discolour the silver strip from a light grey or brown shade to a distinct grey up to black.

12 Expression of results

The result shall be indicated with reference to this standard as follows:

corrosive sulfur as specified in DIN 51 353: absent

or

corrosive sulfur as specified in DIN 51 353: present.

Standards referred to

DIN 12 111	Testing of glass, grain method of determining the water resistance of glass at 98 °C and classification of glass into hydrolytic glasses
DIN 12 242 Part 1	Laboratory glassware; interchangeable conical ground joints; dimensions, tolerances
DIN 12 387	Laboratory glassware; conical flasks with 1 : 10 conical sockets
DIN 50 011 Part 1	Testing of materials, structural components and equipment; heating cabinets; concepts, requirements
DIN 51 750 Part 1	Testing of petroleum products; sampling; general instructions
DIN 51 750 Part 2	Testing of petroleum products; sampling, liquid substances
DIN 53 135	Filter papers for chemical analysis; classification, designation, main properties, methods of test
DIN 61 640	Surgical dressings; absorbent waddings for medical use
DIN 69 176 Part 2	Grains of fused aluminium oxide and silicon carbide for coated abrasives; testing P 12 to P 220 macrogrits
DIN/VDE 0370 Part 1	Insulating oils; fresh insulating oils for transformers, transducers and switchgear
DIN/VDE 0370 Part 2	Insulating oils; used insulating oils in transformers, transducers and switchgear

Other relevant standards

ASTM D 1275-67 *) (Reapproved 1981) Standard test method for corrosive sulfur in electrical insulating oils

Previous editions

DIN 51 353: 01.65, 09.77

Amendments

The following amendments have been made in comparison with the September 1977 edition:

The text of the standard has been editorially revised, taking into account the specifications given in DIN 1310 and the standards of the DIN 820 series; in addition, the "standards referred to" clause has been brought up to standards practice.

Explanatory notes

This standard has been prepared, on the basis of many tests, by Technical Committee *NMP 661 Prüfung von Schmierölen, sonstigen Ölen und Paraffinen* of the *Fachausschuß Mineralöl- und Brennstoffnormung* of the *Normenausschuß Materialprüfung* (Materials Testing Standards Committee). Solutions prepared from elemental sulfur and sulfurized fatty matter in insulating oil were used as reference substances.

According to ASTM method D 1275 - 58, designed for the same purpose, a copper strip was to be kept in oil for 19 hours at 140 °C and the discolouration of the strip indicated by a code number, used for classifying the oils according to their corrosivity. Here, too, repeatability was unsatisfactory and it was difficult to allocate the code numbers, mainly because of the wide variety of colour shades on copper. In the meantime, the above ASTM standard was replaced by D 1275 - 67, which specifies a similar method of test, but provides for a classification only into corrosive and non-corrosive oils.

The V-shape of the silver strip and the preparation of the test sheet have been adopted from the ASTM method. For achieving satisfactory repeatability of test results, it was found to be of major importance to lay down precisely the procedure of preparing the test sheets. The metal surface is highly reactive immediately after the polishing. After storage of 1 minute at 100 °C, the reactivity is reduced to a repeatable degree, but is still sufficient for the unambiguous detection of 2 mg of elemental sulfur in 1 kg of oil. Sheet silver treated with nitric acid, hydrochloric acid, ammonia solution or cyanide proved to be much less sensitive.

International Patent Classification

G 01 N 31/22

*) Obtainable from *Auslandsnormenvermittlung im DIN*, Burggrafenstraße 6, D-1000 Berlin 30.